

Tutorial 5 Projectile Motion Suggested Solutions

D1 (a) (i) Vertically, $s_y = 0$

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$0 = (u \sin \theta) t - \frac{1}{2} g t^2$$

$$t = \frac{2u \sin \theta}{g} \quad \text{OR} \quad 0 \text{ (reject)}$$

(ii) To reach maximum height, $t_H = \frac{1}{2} t = \frac{u \sin \theta}{g}$

$$\begin{aligned} \text{(iii)} \quad R &= u \cos \theta \left(\frac{2u \sin \theta}{g} \right) = \frac{2u^2 \sin \theta \cos \theta}{g} \\ &= \frac{u^2 \sin 2\theta}{g} \end{aligned}$$

(b) For projectile motion, angle of projection $\theta < 90^\circ$,
Maximum $\sin 2\theta = 1$

$$2\theta = 90^\circ$$

$$\therefore \theta = 45^\circ$$

This gives maximum range R .

Or:

To get maximum R

$$\frac{dR}{d\theta} = 0 = \frac{d}{d\theta} \left(\frac{u^2 \sin 2\theta}{g} \right) = \frac{u^2}{g} (2 \cos 2\theta)$$

$$\cos 2\theta = 0 \Rightarrow 2\theta = 90^\circ \quad (\text{Note: can go on to prove } 2^{\text{nd}} \text{ derivative} < 0)$$

$$\therefore \theta = 45^\circ$$

D2 (a) $s_y = +8.0 \text{ m}$, $v_y = 0 \text{ m s}^{-1}$, $u_y = u \sin 50^\circ$, $a_y = -9.81 \text{ m s}^{-2}$

$$v_y^2 = u_y^2 + 2a_y s_y$$

$$0 = (u \sin 50^\circ)^2 + 2(-9.81)(8.0)$$

$$u = 16.35 = 16.4 \text{ m s}^{-1}$$

(b) $s_y = -2.0 \text{ m}$, $u_y = u \sin 50^\circ$, $a_y = -9.81 \text{ m s}^{-2}$

$$v_y^2 = u_y^2 + 2a_y s_y$$

$$v_y^2 = (u \sin 50^\circ)^2 + 2(-9.81)(-2.0)$$

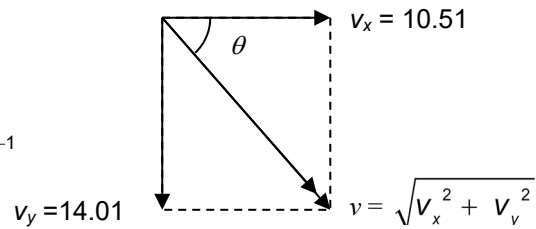
$$v_y = -14.01 = 14.0 \text{ m s}^{-1} \text{ (downwards)}$$

$$v_x = u \cos 50^\circ = 16.35 \cos 50^\circ = 10.51 \text{ m s}^{-1}$$

$$v = \sqrt{v_x^2 + v_y^2} = 17.5 \text{ m s}^{-1}$$

$$\tan \theta = \frac{14.01}{10.51}$$

$$\theta = 53.1^\circ \text{ below the horizontal}$$



(c) $v_y = u_y + a_y t$

$$-14.01 = (16.35 \sin 50^\circ) + (-9.81)t$$

$$t = 2.7 \text{ s}$$

Note: substitute v_y as -14.01 and not $+14.01$ to find the correct time of flight !

D3 (a) Taking downwards as positive

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$600 = (u \cos 60^\circ)(5.0) + \frac{1}{2}(9.81)(5.0)^2$$

$$u = 190.95 = 191 \text{ m s}^{-1} \text{ (3sf)}$$

(b) $s_x = u_x t = (190.95 \sin 60^\circ)(5.0) = 826.83 = 827 \text{ m (3sf)}$

(c) $v_y^2 = u_y^2 + 2a_y s_y = (190.95 \cos 60^\circ)^2 + 2(9.81)(600) = 20887$

$$v_x = u_x = (190.95 \sin 60^\circ)$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(190.95 \sin 60^\circ)^2 + 20887} = 219.62 = 220 \text{ m s}^{-1} \text{ (3sf)}$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{20887}}{190.95 \sin 60^\circ} \right) = 41.2^\circ \text{ below the horizontal}$$

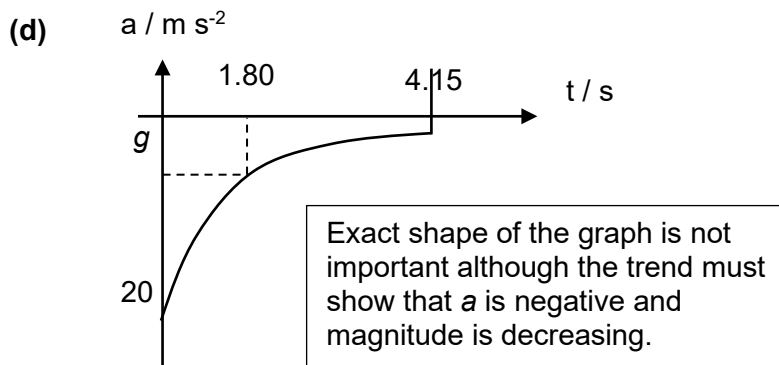
D4 (a) At maximum height, velocity of ball = 0 m s^{-1}
Hence, time = 1.80 s (2 d.p.)

(b) After the ball leaves the thrower's hand, it experiences both the force of gravity (its weight, mg) and drag force (kv) in the downward direction.

Resultant (downward) force on the ball, $ma = mg + kv$
Hence, acceleration of the ball, $a = g + kv/m > 9.81 \text{ m s}^{-2}$

Note: After the ball leaves the thrower's hand, it has a high speed (v). Therefore, the acceleration of the ball would be much larger than g .

(c) 1.80 s .
This is the point when the ball is momentarily at rest. It does not experience any drag force (since $v = 0$) and its resultant force is just its weight. Hence, its acceleration at this point is equal to g .



- (e) From the graph, it can be seen that the change in speed on the way up is 26 m s^{-1} and the change in speed on the way down is 15 m s^{-1} .

This implies that the average speed on the upward journey is greater than that on the downward journey. Since the distance travelled in both directions must be the same, the time taken to reach the maximum height must be shorter than the time taken to return to the starting point.

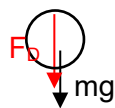
[Recall: Ave speed = Distance travelled / Time taken

⇒ Time taken = Distance travelled / Ave speed

⇒ For the same distance travelled, time taken $\propto (\text{Ave Speed})^{-1}$]

Alternative approach:

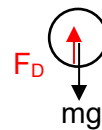
On the way up



$$ma_{\text{up}} = mg + F_D$$

$$a_{\text{up}} = g + F_D/m$$

On the way down



$$ma_{\text{down}} = mg - F_D$$

$$a_{\text{down}} = g - F_D/m$$

Hence, $a_{\text{up}} > a_{\text{down}}$

The ball slows down faster (on the upward journey) than it speeds up (on the downward journey).

⇒ Time taken to reach maximum height < time taken to return to starting point

- D5 (a)** From $t = 0$ to $t = 1$ s, horizontal distance travelled = 4.25 m

From $t = 1$ s to $t = 2$ s, horizontal distance travelled = $8.5 - 4.25 = 4.25$ m

From $t = 2$ s to $t = 3$ s, horizontal distance travelled = $12.75 - 8.5 = 4.25$ m

From $t = 3$ s to $t = 4$ s, horizontal distance travelled = $17 - 12.75 = 4.25$ m

From $t = 4$ s to $t = 5$ s, horizontal distance travelled = $21.25 - 17 = 4.25$ m

From $t = 5$ s to $t = 6$ s, horizontal distance travelled = $25.5 - 21.25 = 4.25$ m

From $t = 6$ s to $t = 7$ s, horizontal distance travelled = $29.75 - 25.5 = 4.25$ m

Since the horizontal distance increases at a constant rate, the horizontal velocity is constant which implies that frictional forces opposing the motion can be taken to be negligible.

(b) $u_x = 4.25 \text{ m s}^{-1}$

(c) $s_y = 40.00 \text{ m}$ (2 d.p.)

(d) (i) $s_y = u_y t + \frac{1}{2} a_y t^2$
 $40 = 0 + \frac{1}{2} a (7.0)^2$

$$a = 1.63 \text{ m s}^{-2}$$

(ii) Loss in $E_p = 0.050 \times 1.63 \times 40$
 $= 3.26 \text{ J}$

(e) Gain in E_K due to vertical motion = Loss in $E_p = 3.26 \text{ J}$

$$E_K \text{ due to horizontal motion} = \frac{1}{2} (0.050) (4.25)^2 = 0.452 \text{ J}$$

$$\text{Total } E_K \text{ after } 7.00 \text{ s} = 3.26 + 0.452 = 3.72 \approx 3.7 \text{ J}$$

(f) $\frac{1}{2} m v^2 = 3.7$

$$v = \sqrt{\frac{2 \times 3.7}{0.050}} = 12.2 \text{ m s}^{-1}$$

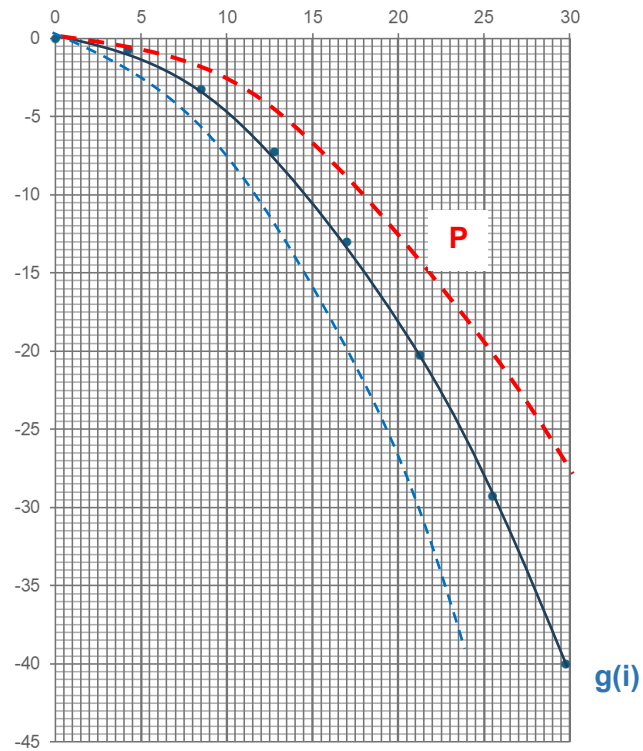
$$v_x = u_x = 4.25 \text{ m s}^{-1}$$

$$\cos \theta = \frac{v_x}{v} = \frac{4.25}{12.2}$$

$$\theta = 69.6^\circ$$

After 7.00 s, the velocity of the sphere is 12.2 m s^{-1} at an angle 69.6° below the horizontal.

(g)



1. Line A : If the Moon had an atmosphere, there would be air resistance on the sphere in both the horizontal and vertical directions.
2. Line P : On a planet with a smaller acceleration due to free fall (but with no atmosphere), the vertical distance travelled per unit time is less than that on the moon ($s = \frac{1}{2}at^2$).

D6 (a) Region A:

When the skydiver just jumps off the helicopter, only gravitational force is significant as the velocity is approximately zero.

$$F_{net} = W = mg$$

Region A corresponds to this instant ($t = 0$) where the skydiver accelerates downward with acceleration equals to g due to the absence of air resistance.

Region B:

As the speed increases, the air resistance F_{air} becomes more significant and increases.

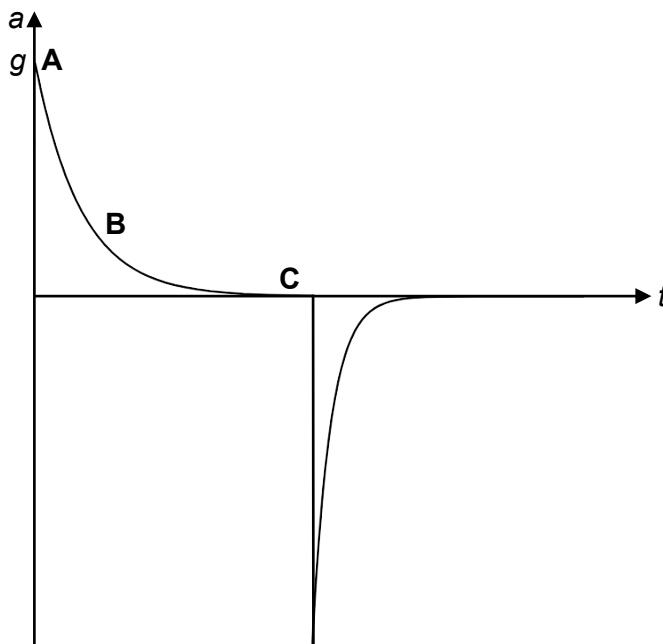
$$F_{net} = W - F_{air} = ma$$

Resultant downward force decreases in magnitude, acceleration decreases. Hence, Region B shows a decreasing gradient (ie speed is increasing at a decreasing rate).

Region C:

When the speed increases to a value such that the magnitude of F_{air} equals the gravitational force, there is zero resultant force and zero acceleration. Hence, at Region C, the gradient of the graph is zero.

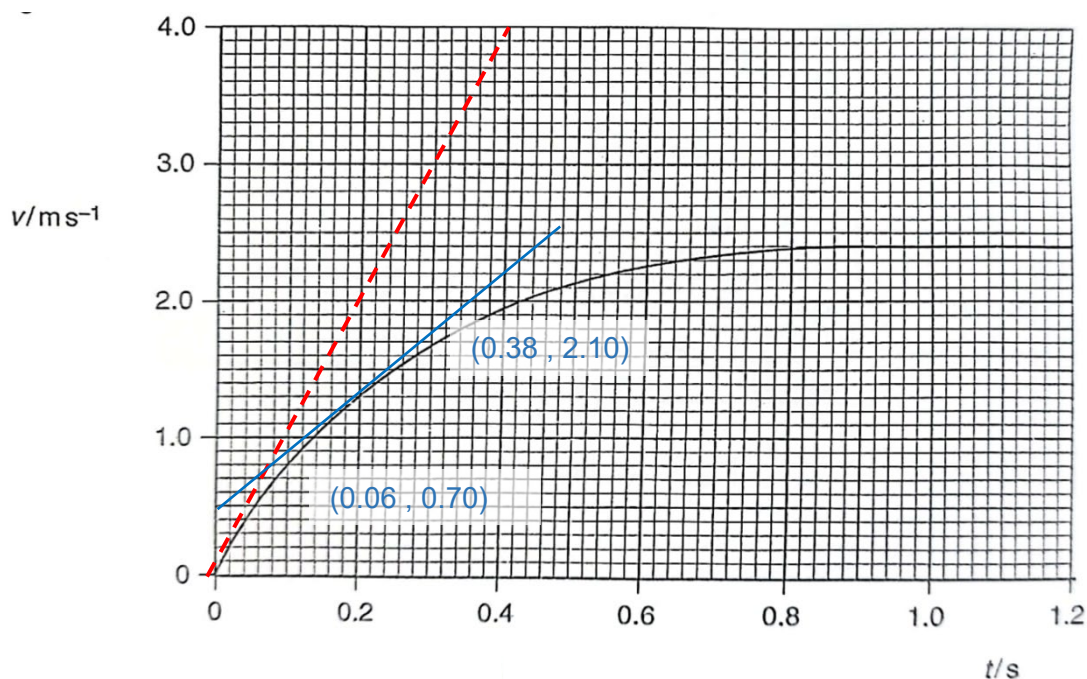
(b)



- D7 (a) (i)** In the absence of air resistance, the horizontal component of velocity remains constant throughout.
- (ii)** In the absence of air resistance, the object experiences constant acceleration due to gravity and the vertical component of its velocity increases at a constant rate.
- (b) (i)** The presence of air resistance will cause the horizontal component of velocity to decrease at a decreasing rate.

- (ii) The presence of air resistance will cause the net force on the object to decrease. Hence, it will move with decreasing acceleration and its vertical component of velocity will increase at a decreasing rate.

D8 (a)



- (b) Gradient of tangent at any point = instantaneous acceleration

$$\text{When } t = 0.20 \text{ s, } a = \frac{(2.10 - 0.70)}{(0.38 - 0.06)} = \frac{1.40}{0.32} = 4.38 \text{ m s}^{-2}$$

- (c) (i) Maximum resistive force = Weight of the ball
 $= 0.015 \times 9.81$
 $= 0.147 \text{ N}$

- (ii) At $t = 0.20 \text{ s}$,

$$\begin{aligned} F_{\text{net}} &= W - F_{\text{air}} = ma \\ F_{\text{air}} &= mg - ma \\ &= 0.015(9.81 - 4.38) \\ &= 0.081 \text{ N (close to } 0.083 \text{ N)} \end{aligned}$$

- (d) When $F_1 = 0.147 \text{ N}$, $v_1 = 2.40 \text{ m s}^{-1}$
When $F_2 = 0.083 \text{ N}$, $v_2 = 1.30 \text{ m s}^{-1}$

If $F \propto v$, then $\frac{F}{v}$ would be a constant value

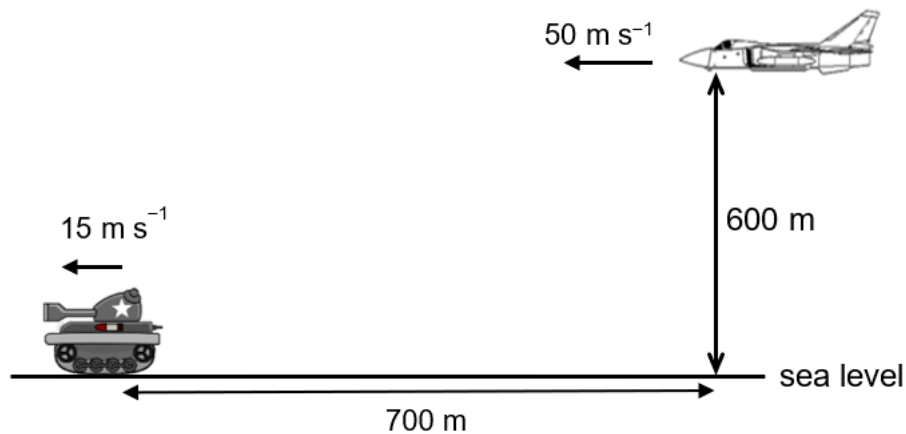
$$\frac{F_1}{v_1} = \frac{0.147}{2.40} = 0.0613$$

$$\frac{F_2}{v_2} = \frac{0.083}{1.30} = 0.0638$$

Since $\frac{F_1}{v_1} \approx \frac{F_2}{v_2}$, the magnitude of air resistance force is proportional to the speed of the ball.

[N12/3/1]

C1



$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$600 = 0 + \frac{1}{2} (9.81) t^2$$

$$t = 11.1 \text{ s}$$

Relative speed of fighter bomber w.r.t tank = $50 - 15 = 35 \text{ m s}^{-1}$

Time needed to cover 700 m horizontal distance is 20 s

$$\therefore 20 - 11.1 = 8.9 \text{ s}$$

C2 Resolve the vectors in the directions parallel and perpendicular to the plane.

$$a_{\parallel} = -g \sin \theta \quad ; \quad a_{\perp} = -g \cos \theta$$

$$s_{\parallel} = u (\cos \phi) t - \frac{1}{2} g (\sin \theta) t^2 \quad \dots\dots\dots(1)$$

$$s_{\perp} = u (\sin \phi) t - \frac{1}{2} g (\cos \theta) t^2 \quad \dots\dots\dots(2)$$

At the point of impact, $s_{\perp} = 0$

Hence, (2) $\Rightarrow t = \frac{2u \sin \phi}{g \cos \theta} \quad \dots\dots\dots(3)$

Substitute equation (3) into (1),

$$\begin{aligned} s_{\parallel} &= u (\cos \phi) \left(\frac{2u \sin \phi}{g \cos \theta} \right) - \frac{1}{2} g \sin \theta \left(\frac{2u \sin \phi}{g \cos \theta} \right)^2 \\ &= \left(\frac{u^2 \sin 2\phi}{g \cos \theta} \right) - \frac{2u^2}{g} \tan \theta \sec \theta \sin^2 \phi \end{aligned}$$

$$\frac{ds_{\parallel}}{d\phi} = \left(\frac{2u^2}{g \cos \theta} \right) (\cos 2\phi) - \frac{2u^2}{g} \tan \theta \sec \theta \sin 2\phi$$

$$\begin{aligned} \frac{ds_{\parallel}}{d\phi} = 0 &\Rightarrow \frac{2u^2}{g} \sec \theta [\cos 2\phi - \tan \theta \sin 2\phi] = 0 \\ &\Rightarrow [\cos 2\phi - \tan \theta \sin 2\phi] = 0 \\ &\Rightarrow \tan 2\phi = \cot \theta \end{aligned}$$

Recall Trigo identity: $\tan \phi = \cot (\pi/2 - \phi)$

$$\begin{aligned} &\Rightarrow \cot (\pi/2 - 2\phi) = \cot \theta \\ &\Rightarrow 2\phi = \pi/2 - \theta \\ &\Rightarrow \phi = \pi/4 - \theta/2 \end{aligned}$$